Interim Report: Change Point Analysis of Brent Oil Prices

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1. Business Objective

Birhan Energies is a consultancy specializing in data-driven insights for the energy sector. This project focuses on **analyzing how political and economic events affect Brent oil prices**, providing actionable intelligence to key stakeholders:

- **Investors:** Need insights to manage risk and optimize returns.
- **Policymakers:** Require accurate analyses to ensure energy security and economic stability.
- **Energy Companies:** Depend on price forecasts for operational planning and supply chain management.

The analysis targets structural changes in daily Brent oil prices caused by major events such as political decisions, conflicts in oil-producing regions, international sanctions, and OPEC policy shifts. Understanding these dynamics supports informed **investment strategies, policy development, and operational planning** in a highly volatile global energy market.

2. Discussion of Completed Work and Initial Analysis (Task 1)

2.1 Data Analysis Workflow

The analysis follows these structured steps:

1. **Data Loading & Preparation:** Import historical Brent oil prices (May 1987 – Sep 2022) and convert dates to datetime format.
2. **Exploratory Data Analysis (EDA):** Examine trends, stationarity, and volatility patterns; compute log returns for improved stationarity assessment.
3. **Event Research:** Compile major geopolitical, economic, and OPEC-related events in a structured dataset.
4. **Model Understanding:** Study Bayesian change point models, including assumptions, expected outputs, and limitations.
5. **Insight Generation:** Identify potential correlations between events and detected change points.

2.2 Event Research

A structured dataset has been prepared with 10–15 major events, including political conflicts, OPEC decisions, and economic sanctions.

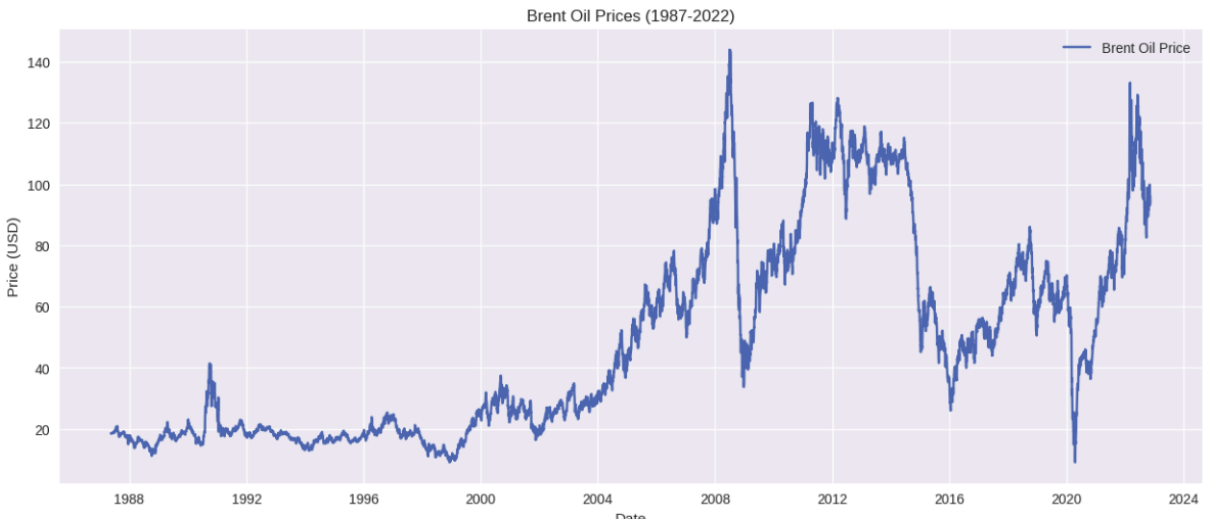
Date	Event	Type	Description
1990-08-02	Gulf War	Conflict	Iraq invades Kuwait, leading to a spike in oil...
1997-07-01	Asian Financial Crisis	Economic	Financial crisis starting in Thailand affectin...
2001-09-11	9/11 Attacks	Geopolitical	Terrorist attacks in the US causing market unc...
2003-03-20	US Invasion of Iraq	Conflict	Concerns over Middle East stability and oil su...
2008-09-15	Global Financial Crisis	Economic	Collapse of Lehman Brothers precipitating a gl...
2010-12-17	Arab Spring	Political	Wave of protests and revolutions in the Arab w...
2011-02-15	Libyan Civil War	Conflict	Uprising against Muammar Gaddafi leading to si...
2014-06-20	2014 Oil Price Crash	Economic/Policy	Surge in US shale production combined with OPE...
2015-07-14	Iran Nuclear Deal (JCPOA)	Political	Agreement lifting sanctions on Iran, allowing ...
2016-12-10	OPEC+ Formation	Policy	OPEC and non-OPEC members (led by Russia) agre...
2018-07-06	US-China Trade War	Economic	Tariffs and trade tensions dampening global ec...
2020-03-08	Russia-Saudi Price War	Policy	Brief price war after disagreement on producti...
2020-03-11	COVID-19 Pandemic	Health/Economic	WHO declares pandemic; global lockdowns cause ...
2021-01-01	Post-COVID Recovery	Economic	Global economies reopening leading to demand o...
2022-02-24	Russia-Ukraine War	Conflict	Invasion of Ukraine leading to sanctions on Ru...

2.3 Assumptions and Limitations

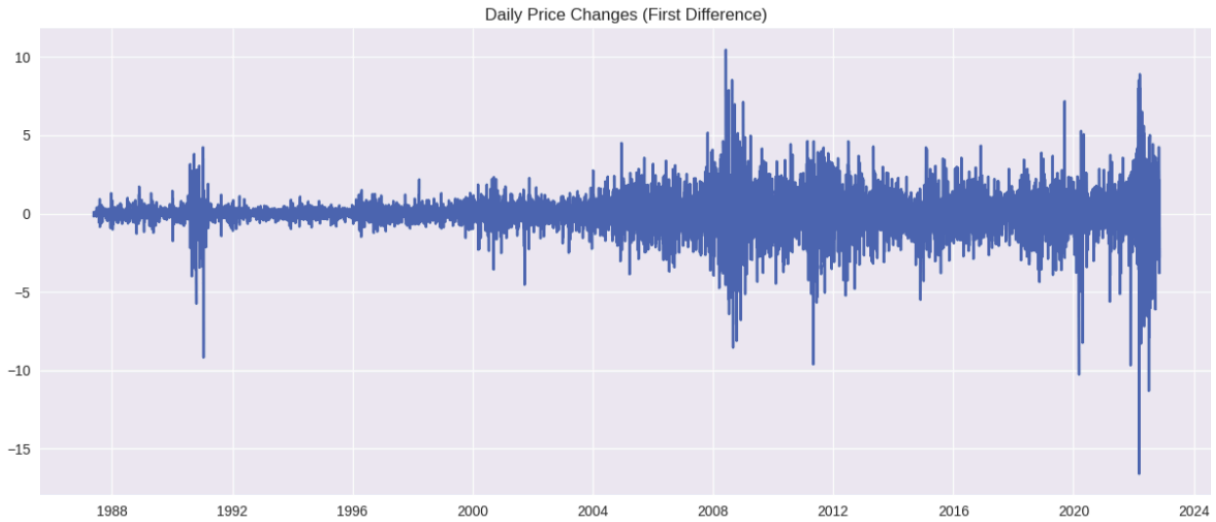
- **Correlation vs. Causation:** Change points indicate statistical shifts but do not prove causal impact.
- **Data Coverage:** Only major documented events are included; smaller events may be excluded.
- **Stationarity & Volatility:** Assumes log returns approximate stationarity; high volatility periods may affect model sensitivity.

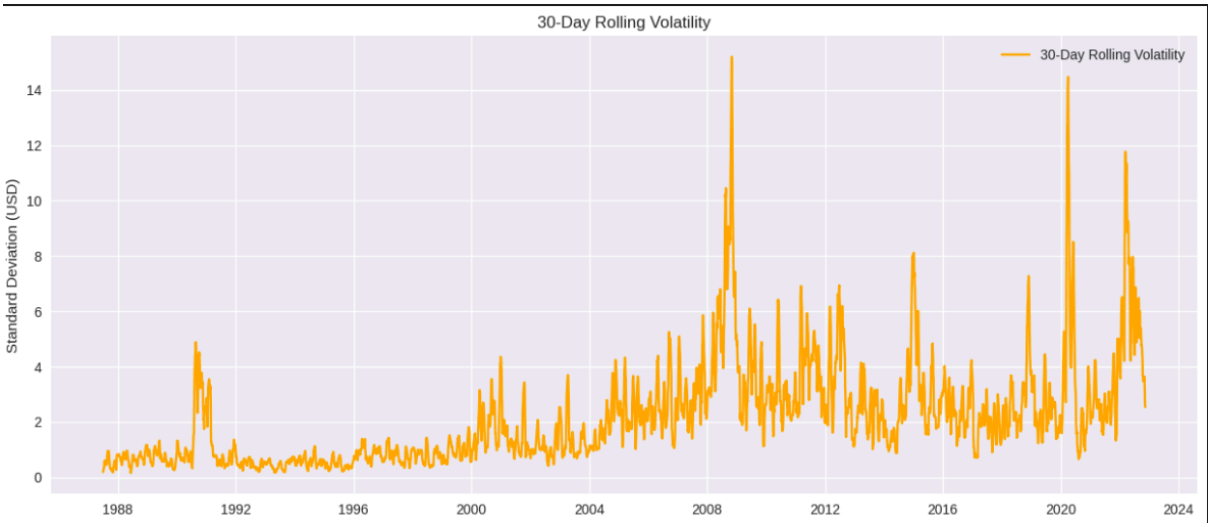
2.4 Initial Time Series Analysis

- **Trend Analysis:** Brent oil prices exhibit long-term upward trends with intermittent spikes and drops.
- **Volatility Patterns:** Notable volatility clustering occurs around major geopolitical and economic events.
- **Stationarity Testing:** Preliminary log return analysis suggests approximate stationarity for modeling purposes.



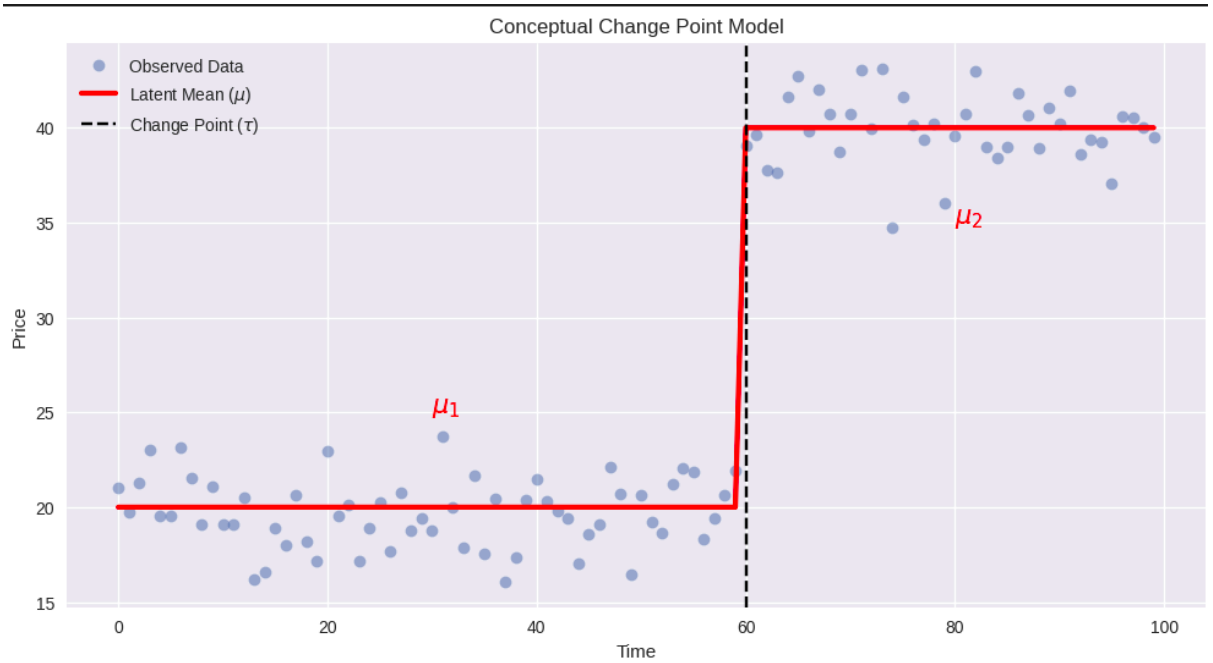

```
--- ADF Test for Raw Prices ---
ADF Statistic: -1.993856
p-value: 0.289274
Result: Non-Stationary (Fail to reject H0)
--- ADF Test for First Difference ---
ADF Statistic: -1.993856
p-value: 0.289274
Result: Non-Stationary (Fail to reject H0)
--- ADF Test for First Difference ---
ADF Statistic: -15.529368
p-value: 0.000000
Result: Stationary (Reject H0)
ADF Statistic: -15.529368
p-value: 0.000000
Result: Stationary (Reject H0)
```





2.5 Change Point Model Understanding

- **Purpose:** Detect structural breaks in oil prices to identify periods of significant change.
- **Model Components:**
  - Switch point (`tau`) as discrete uniform prior
  - Before/after parameters (μ_1, μ_2)
  - Switch function to assign parameters based on time index
- **Expected Outputs:**
  - Posterior distribution of change points
  - Estimates of mean/variance before and after detected points



3. Next Steps and Key Areas of Focus

3.1 Task 2 – Bayesian Change Point Modeling (PyMC)

- Implement full Bayesian model for Brent prices.
- Run MCMC sampling and check convergence.
- Identify and quantify structural breaks in the time series.
- Associate detected change points with compiled events and provide **quantified impacts**.

3.2 Task 3 – Dashboard Development

- **Backend (Flask):** Serve historical data, model outputs, and event correlations through API endpoints.
- **Frontend (React):** Interactive visualizations including event highlights, date filters, and drill-down functionality.
- **Outcome:** Stakeholders can efficiently explore how specific events influence Brent oil prices.

4. Deliverables for Interim Submission

1. Structured CSV of key events (10–15 events).
2. Initial EDA visualizations (price trends, log returns, volatility).
3. Documentation of assumptions, workflow, and preliminary findings.

5. References

- PyMC Documentation: https://docs.pymc.io
- Brent Oil Price Dataset: [Dataset source URL]
- Bayesian Change Point Literature: [Relevant papers / URLs]